

BEE 208 — Question 7: Total Power of Lighting Points

Control Statement: for loop | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables wattage, totalPower (double) and i (int)

Step 3: Set totalPower = 0

Step 4: Use a for loop from i = 1 to 12

Step 5: Prompt user to enter wattage for lighting point i

Step 6: Read wattage

Step 7: totalPower = totalPower + wattage

Step 8: End loop

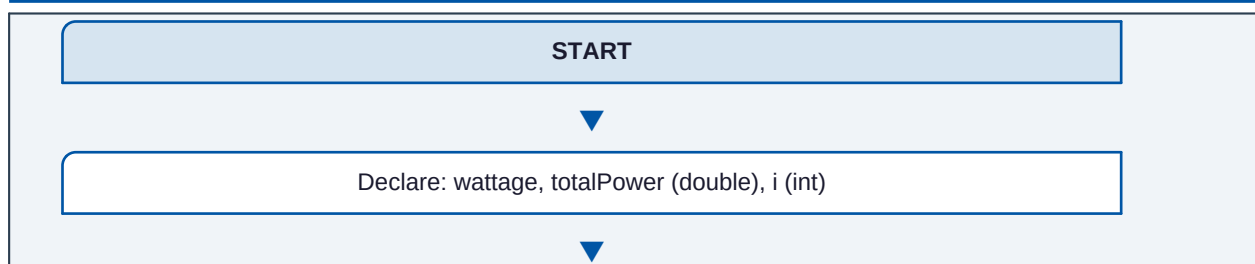
Step 9: Display totalPower in watts

Step 10: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE wattage, totalPower AS DOUBLE  
  
DECLARE i AS INTEGER  
  
SET totalPower = 0  
  
FOR i = 1 TO 12 DO  
  
    OUTPUT 'Enter wattage for Lighting Point', i, ':'  
  
    INPUT wattage  
  
    totalPower = totalPower + wattage  
  
END FOR  
  
OUTPUT 'Total Lighting Load:', totalPower, 'W'  
  
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	12 readings: all 60 W	Expected Output	Total = 720 W
Input	12 readings: all 100 W	Expected Output	Total = 1200 W